

Optimal EEG Feature Extraction based on R-square Coefficients for Motor Imagery BCI System

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Abstract: In cue based motor imagery brain computer interface (BCI) paradigm, subject was stimulated by different cue to distinguish different imagination task and proceed by imagination tasks. These evoked the specific related frequency rhythms. But the problem appears as these rhythms change due subject and imagination task. While some subject is fast induce the electrical signal rhythms, other has long latency in induce them. This related time temporal problem of imagination. To boost the accuracy of EEG signal translation for reliable system, in this paper, we proposed a method for extracting optimal feature. First, EEG signals were extracted and apply to Laplace filter and band pass filter with pass band frequency of 7Hz to 40Hz. Channels C3, Cz, and C4 were applied to Short-Time Fourier Transform with frequency band of 1Hz. R-square correlation coefficient of three channels were found and selected the best frequency and time parameter. Finally, with the selected parameters, optimal STFT feature extracted. We simulated the classification accuracy with linear discriminant analysis with BCI competition IV, III and our laboratory dataset.

Keywords: Optimal Feature Extraction, R-square Coefficient, Motor Imagery, Electroencelelography, Brain-computer Interface.

1. INTRODUCTION

In last few decades, brain computer interface (BCI) system has captured many researchers intention. This system provides a new communication channel directly from the brain to the application device start from the word spelling to driving the wheel chair[1]. Many researchers have proved that with invasion technique it is possible for disable patience to controlling external environment and device with high accuracy rate in command recognition using the imagination of motor movement like left hand, right hand and foot movement.

Moving forward, researchers and engineers intend to use this system for not only patient but also normal people for basic daily life application routine. Instead of invasion method, electroencelelography (EEG) signals were used. This method composed of recording the EEG signals from brain via electrodes in a scalp that contact with skull. And these electrical signals were digitize and apply into feature extraction and classification methods before translated into command control for BCI application[2].

To extract the most relevant information from the EEG signal is a difficult task. EEG single natural has small resolution in micro-voltage that easily contaminated with noise from nearby electrodes and other source resulting low signal to noise ratio. More the subject evoked potential itself plays the role of contaminated the signal. Event related potential was found related to input stimuli or user intention. While the most researchers used event related desynchronized and synchronized (ERD/ERS) to recognize the subject imagination discriminant related to motor imagery task. Separation these two events could enhancing the signal to noise ratio of that EEG signal[3].

ERD /ERS itself is non-time lock event. In BCI cue based motor imagery system (synchronous), these two

events happened in few seconds after the subject was given corresponding stimuli. In contrast, for asynchronous system, subject started imagination of motor task without specific stimuli. The system need to be intelligence enough to recognize the activation of motor imagery. To solve this problem, the epoch size for during that time was selected to analyze, resulting large time sample that does not hold the relevant in and cause the trade off the accuracy of classification prediction. The short temporal information need to from the EEG signal.

In[4], [5] and [6], the study had showed that spectral component of brain activation related to motor imagery task using ERD found in μ band(10-12Hz) and ERS found in beta band(13 -30Hz), but they vary from subject to subject significantly. To get the optimal features over that frequency band, frequency selection method like in [7] for subject specific frequency band need to be search and selected independently from each subject.

In [8], spatial information had been consider. Even though neurophysiology confirm that the most significant channel location related to motor imagery are C_3 , C_4 and C_z , the EEG signals are spread over the area around these electrode. In different experiment and different subject the distribution of EEG signal over the location is different. To solve this, channel selection methods were proposed [9].

To adapt BCI system to be used wide, the spatial-spectral-temporal parameter of EEG signal need to be search and tuned before using in the system. In [10] spatio-spectral and temporal parameter searching algorithm using class correlation and particle swarm was proposed. This proposed method composes of series of parameter searching in order of channel, time and frequency band of EEG. In this process, show rejection concept of the focus ERD/ surrounded ERS

which can be used in motor imagery task. But it showed the promising result and confirmed that these specific parameters need to analysis for real life BCI application.

In [11], author proposed using genetic algorithm (GA) for automatic feature extraction in P300 detection. The variation of coding the information of gene in a chromosome and natural selection of GA could be used in searching algorithm for optimal feature based on the spatio-spectral-temporal parameter.

In this paper, we proposed a method for optimal feature time-frequency feature extraction subject parameter searching.

The remains of this paper are organized into four sections. Section 2 contains details dataset from the BCI competition IV and our laboratory dataset. In section 3, we give the detail our proposed feature extraction method. Section 4 covers on the experimental and results. In the last section, we draw the discussion and conclusion over our research.

2. EXPERIMENTAL PARADIGM

2.1 BCI competition IV and III

Dataset 1 of BCI competition IV provided by Berlin BCI (BBCI) group were used in the experiment. . In the experiment, subject was asked to sit in the arm chair in relax position in front of the screen. In the first 2 seconds, blank screen was shown on the screen. At the time 2s, a cross section was shown on the screen for 2s to alert the subject of next coming trial. At time of 4s a cue of right pointed arrow or left arrow appeared on the screen and subject performs the imagination task of left or right hand movement. The cross section is superimposed on the cue. Totally, each subject performed 200 trials of imaginations. The dataset were sampled with frequency of 100Hz and 59 channels of standard 10-20 system were extracted. There are totally 7 subjects named “a, b, c, d, e, f, g”, which only four subjects “a, b, f, g” is real subjects, while the other were artificial generated. In our simulation, we selected only subject “b, g” which performed task of imagination of left and right hand movement.

Also the BCI completion III, dataset IVa, is used in the simulation. The experiment paradigm is showed in the Fig. 1. There is no fixation cross in this experiment and tasks are self-imagination of right hand and right foot movement. Dataset sampling rate is 100Hz, consist of 280 trial and 5 subjects.

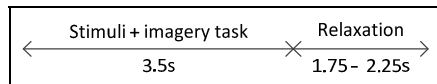


Fig. 1 BCI competition III dataset IVa: Time scheme in calibration session. The epoch time is from 4.25-5.75s.

2.2 Motor imagery EEGs acquitting

In our laboratory, we did experiment of motor imagery task of left and right hand movements. The experiment, the cue-based BCI paradigm with no

feedback, consisted of two different motor imagery tasks: the imagination of left hand movement (class 1) and right hand movement (class 2). One session was conducted for each subject, where one session consists of 5 run and 40 trials for each run. There is a short break between each run. Totally, there are 200 trials for one session.

Figure 2 shows time scheme of our experiment. The subject was sitting in a comfortable armchair in front of a computer screen at distance about 1m. At the beginning of trial ($t = 0s$), a white fixation cross appear on the black screen to warn the subject to prepare for the upcoming trial. The fixation displayed on the screen for 2 seconds. Then at time $t = 2s$, a cue appeared on the screen. The cue displayed random pattern where white arrow pointed to left of screen corresponds to imagination of left hand movement task and the right pointed arrow to right corresponds to imagination of right hand movement. After cue appeared, subject saw the arrow and thought of word “Left” or “Right” corresponding to the direction of arrow. At time $t = 4s$, a beep was played and a white ball appeared on the screen to show for subject start the imagination task. During imagination task, the subject tried to imagine using corresponded arm clutching the ball by closing and opening hand. In 4s of imagination task, subject tried to attempts 1 task of imagination of opening and closing. At time $t = 8s$, blank screen appeared determining the end of task. The blank screen appeared for 2 second before start of next trail.

EEG signals were measure by NeuroScan 4.5 with 64 channels EEG using the 10-20 system quick caps setting. All channels were recorded monopolarly with the left mastoid serving as the reference ground located at channel AF_2 . The signals are sampled with 500Hz and band pass-filter between 0.5Hz to 100Hz. And a notch filter at frequency at 60Hz is enabled to suppress the line power noise. We experiment with totally 5 subjects.

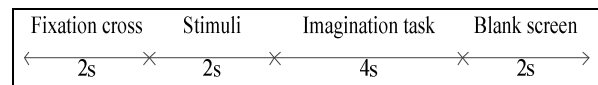


Fig. 2 URIS dataset: left and right hand motor imagery experimental paradigm with epoch size of 10s

3. OPTIMAL FEATURE EXTRACTION BASED ON R-SQUARE COEFFICIENTS

We proposed a feature extraction method composed of several step as indicate in Fig. 3. First, EEG signals from all channels are filtered Laplace and band pass filter. Only channels C_3 and C_4 are selected for analysis. STFT is used to transform EEG signals into time-frequency features. The features are select for the optimal parameter by R-square coefficient and. The optimal features obtain by using the optimal spectral and time parameters. At last stage, the optimal feature is classifier by linear discriminant analysis to prove the improvement of feature extraction method.

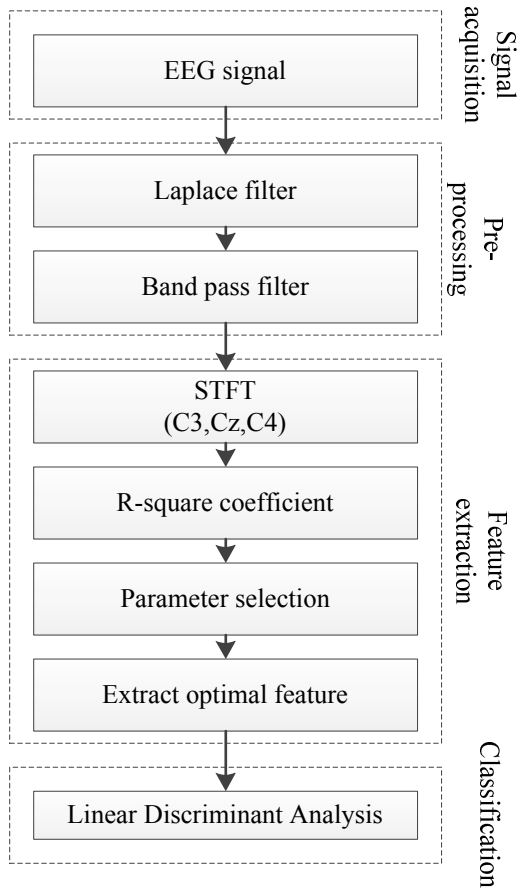


Fig. 3 Optimal feature extraction based R-square coefficient method

3.1 Preprocessing

Laplacian spatial filter is comprised of discretized approximation of the second spatial derivation of two-dimensional Gaussian distribution on the scalp surface, and attempts to invert the process that blurred the brain signals detected on the scalp. The approximations are simplified such that, at each time point t , the weighted sum of the potential s_i of the four nearest or next nearest electrodes is subtracted from the potential s_h at a center electrode for the small and large Laplacian, respectively given in Eq. (1). Eq. (2) is the weight $w_{h,i}$ as a function of the distance $d_{h,i}$ between the electrode of interest h and its neighbor i in set S_i . In our work we chose Euclidian distance.

$$s'_h(t) = s_h(t) - \sum_{i \in S_i} w_{h,i} s_i(t) \quad (1)$$

$$w_{h,i} = \frac{1}{\sum_{i \in S_i} \frac{1}{d_{h,i}}} \quad (2)$$

In EEG based motor imagery, the useful frequency rhythm that contain informative feature in α and β band. The other frequency bands consider being less

informative and easy to cause the tradeoff accuracy of classifier[12]. In our simulation, elliptical infinite impulse response (IIR) filter of frequency band of 8Hz to 40Hz was applied signals.

3.2 Optimal EEG feature extraction

The Fourier transform summarizes the similarity between a signal and a sinusoid with a single complex number. The magnitude of the complex number captures the degree to which oscillations at a particular frequency contribute to the signal's energy, while the argument of the complex number captures phase information. Short-time Fourier transform (STFT) is a technique applying a chosen time function or window to the input signal and calculate the Fourier transform. The mathematical properties describe by the Eq. (3). $x(t)$ is the input signal and $w(t)$ is the window function and $X(\omega, t)$ is the transformed signal as a function of normalized frequency ω and time τ .

$$X(\omega, \tau) = \int x(t)w(t - \tau)e^{-j\omega\tau} dt \quad (3)$$

In our simulation we apply the hamming window into STFT with frequency band of 2 Hz range from 0 Hz to 40Hz. Totally we got 21 frequency band components.

R-square coefficient is a square of Pearson correlation coefficient of two random samples x_i and y_i . It is given in Eq. (4). $\bar{x}$, $\bar{y}$ are means of sample x and y , n is total number of sample. In our work sample x_i and y_i are STFT sample from channel C_3 and C_4 of the EEG signal. We find R_t^2 component related to time component and R_f^2 component related to frequency component.

$$R^2 = \frac{\left(\sum x_i y_i - n \bar{x} \bar{y}\right)^2}{\left(\sum x_i^2 - n \bar{x}^2\right)\left(\sum y_i^2 - n \bar{y}^2\right)} \quad (4)$$

To obtain optimal frequency parameter of separation between signals from each class, we select value of frequency as in Eq. (5).

$$f_{optimal} = \arg_f \max(R_f) \quad (5)$$

While the optimal temporal parameter is chosen by Eq. (6) and select the total ranges 0.25s around the optimal value t_{best} as in Eq. (7).

$$t_{optimal} = \arg_t \max(R_t) \quad (6)$$

$$T_{optimal} = [t_{optimal} - 0.25 \quad t_{optimal} + 0.25] \quad (7)$$

After optimal spectral parameter $f_{optimal}$ and temporal parameter $T_{optimal}$ have been found, we applied this parameter for selecting the optimal time-frequency from both channel C_3 and C_4 .

4. EXPERIMENTS AND RESULTS

We applied dataset 1 of the BCI competition IV for analysis. We simulated the result of R-square coefficient correspond to class of hands movement imagination.

Fig. 4 shows the head topo-frequency range from 8 to 24Hz of most active R-square coefficient correspond subject imagined moving right hand of subject "b". While the color bar show the value of R-square coefficient. Among these frequency bands, the most different between channels C_3 and C_4 is at 16Hz.

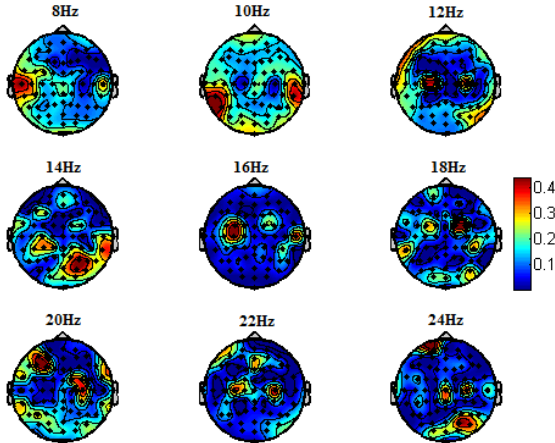


Fig. 4 Topo-frequency of head model correspond to right hand movement

We select the topo-frequency at 16Hz for and show in Fig. 5. It shows clearly that, area around channel C_3 is the most active compare to area around channel C_4 while subject imagine moving right hand and the most active frequency for subject "b" is 16Hz.

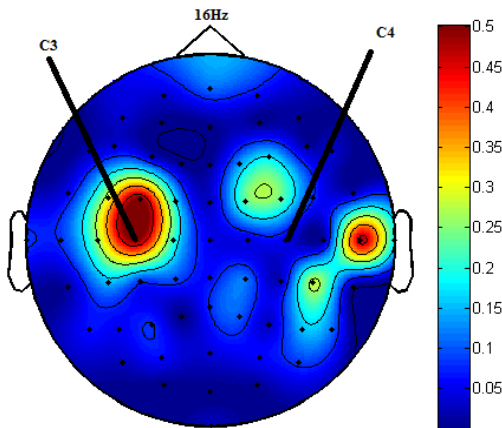


Fig. 5 Topo-frequency at 16Hz head model correspond to right hand movement

Fig. 6 shows the head topo-frequency range from 8 to 24Hz of most active R-square coefficient correspond subject imagined moving left hand of subject "b". While the color bar show the value of R-square coefficient. Among these frequency bands, the most different between channels C_3 and C_4 is at 20Hz.

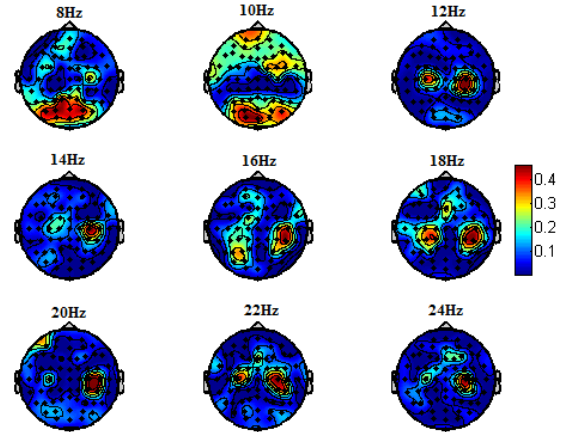


Fig. 6 Topo-frequency of head model correspond to left hand movement

Topo-frequency at 20Hz shows in Fig. 7. It shows clearly that, area around channel C_4 is the most active compare to area around channel C_3 while subject imagine moving left hand and the most active frequency for subject "b" is 20Hz.

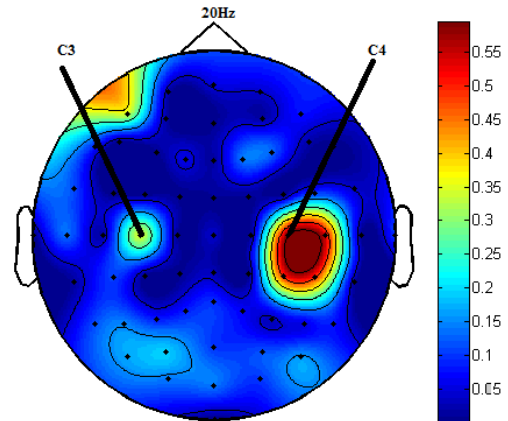


Fig. 7 Topo-frequency at 20Hz head model correspond to left hand movement

Fig. 5 and 7 show even concrete evident over the neurophysiology study that the optimal frequency changing from subject to subject, and more in the imagination task. While the time course related to time onset of imagination of the subject.

5. CONCLUSION

We conclude that time onset of most active feature is changing from one trial to another. Meanwhile the active frequency band is changing from subject to subject and even from a class to another class of imagination movement. A BCI base fixed channel location may result in low accuracy because of EEG signal is distributed over an areas rather than a fix position. While multi-channel BCI system could effective by the artifact from many channel.

To improve BCI performance, searching method for selecting optimal features and signal enhancement

method over the area of analysis electrode should be consider. Yet our research still in progress, we will investigate the performance of our proposed method and improve this system to be available for online BCI.

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